

Case Study: Protecting Digital Intellectual Property Using IP India and I4C

Intellectual Property Rights and Emerging Digital Technologies – I4C + IP India

Subject: Information Security / Cybersecurity

Topic: Intellectual Property Rights and Emerging Digital Technologies

Mode: Individual Self-Learning

Time: 40 Minutes

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Official Resources

IP India: [Intellectual Property India – Government of India](#)

I4C: [Indian Cyber Crime Coordination Centre – Ministry of Home Affairs](#)

Case Study: Protecting Digital Intellectual Property Using IP India and I4C

Case Background

Consider an Indian technology startup named **CyberAI Solutions Pvt. Ltd.**

The company develops AI-based cybersecurity software. During its development, the company creates several valuable digital assets:

- An original AI software program
- A unique technical process for detecting cyber threats
- Company name and logo
- Original website content and training materials
- Proprietary AI models
- Confidential datasets
- Source code
- Business strategies

The company discovers the following incidents:

Incident 1 – Software Copying

A competitor copies portions of the company's software source code and incorporates them into its own commercial application.

Incident 2 – Technology Replication

The competitor develops a similar technical solution using knowledge obtained from CyberAI's confidential research.

Incident 3 – Brand Misuse

Another website uses a logo and company name that are deceptively similar to CyberAI Solutions.

Incident 4 – Digital Content Theft

The company's website articles, training videos and technical documentation are copied and published on another website without permission.

Incident 5 – Cyber Attack

An employee's account is compromised through phishing. The attacker accesses confidential source code and downloads the company's proprietary AI dataset.

The company now has two major questions:

How can IP India help protect the company's Intellectual Property?

and

How can I4C and cybersecurity mechanisms help when the company's digital assets are attacked or misused?

Task 1 – Understand IPR

Using the official **IP India** resources, explain:

1. What is Intellectual Property?
2. What are Intellectual Property Rights?
3. Why are IPR important for technology companies?
4. Why is digital intellectual property difficult to protect?
5. What is the difference between Copyright, Patent, Trademark and Trade Secret?

IP India's official resources identify patents, designs, trademarks and copyright as important categories of intellectual property protection. ([IP India](#))

Task 2 – Identify the Correct IPR

Analyse the CyberAI Solutions case.

Company Asset	Appropriate IPR / Protection	Reason
Source code		
AI technical invention		

Company logo		
Company name		
Website articles		
Training videos		
Proprietary algorithm		
Confidential dataset		
Business strategy		

Student Challenge

For every asset, explain:

Asset → IPR → Reason → Protection Method

Task 3 – Copyright Analysis

CyberAI's software code, website articles, training videos and technical documentation have been copied.

Answer:

1. Which assets may involve copyright protection?
2. Why is software relevant to copyright?
3. What evidence should CyberAI preserve?
4. How can the company detect unauthorised copying?
5. What cybersecurity controls can reduce unauthorised access?

Students should refer to the **Copyright** resources available through IP India's official website. ([IP India](#))

Task 4 – Patent Analysis

CyberAI has developed a new technical process for detecting cyber threats.

IP India's guidance explains that a patent may be granted for an invention that is a **new product or process**, involves an **inventive step**, and is **capable of industrial application**. ([IP India](#))

Students answer:

1. What is a patent?

2. What is an invention?
3. What is novelty?
4. What is an inventive step?
5. What is industrial applicability?
6. Could CyberAI's technical invention potentially require patent evaluation?
7. Why should a company consider patent protection before publicly disclosing an invention?

Task 5 – Trademark Analysis

A competitor uses a logo and company name very similar to **CyberAI Solutions**.

According to IP India's trademark guidance, a trademark distinguishes the goods or services of one entity from those of others and helps protect brand identity and goodwill. ([IP India](#))

Answer:

1. What is a trademark?
2. Why is the company logo important?
3. Why can a similar logo create customer confusion?
4. How can CyberAI search existing trademarks?
5. Why should a technology startup register its brand?

Students should explore IP India's **Trademark Public Search** facility. IP India also provides AI-based trademark search functionality. ([IP India](#))

Task 6 – Trade Secret + Cybersecurity

CyberAI's proprietary dataset, source code and business strategy are confidential.

Students must determine:

Which information should remain confidential?

- Source code
- Passwords
- AI model parameters
- Proprietary datasets
- Customer information
- Business strategy
- Research documents

Then answer:

How can cybersecurity protect confidential intellectual property?

Consider:

- Access control
- MFA
- Encryption
- Least privilege
- DLP
- Logging
- Backup
- NDA
- Employee awareness
- Endpoint security

Task 7 – I4C Connection

Now analyse the **cybercrime component** of the case.

The phishing attack resulted in unauthorised access to CyberAI's source code and confidential dataset.

Students should investigate the official **I4C** website and answer:

1. What is I4C?
2. Why is I4C important for India's cybercrime ecosystem?
3. How can cybercrime reporting help organisations?
4. What type of digital evidence should CyberAI preserve?
5. Why is coordination between organisations, financial institutions, telecom operators and law-enforcement agencies important?
6. How can cybercrime intelligence help identify attackers or related infrastructure?

Important distinction:

IP India primarily addresses the legal/administrative protection and registration ecosystem for IPR, while I4C is relevant to the cybercrime and cybersecurity response ecosystem.

Students should not treat I4C as an IPR registration authority.

Task 8 – IP India vs I4C

Complete this table.

Situation	IP India	I4C / Cybercrime Ecosystem
Patent protection		

Trademark registration		
Copyright protection		
Software/code copying		
Phishing attack		
Hacking of source code		
Cybercrime reporting		
Digital evidence		
Brand misuse		
Protection of confidential digital assets		

Challenge

Identify **three situations where both IPR protection and cybersecurity controls are required simultaneously.**

Task 9 – Emerging Digital Technologies

Analyse the impact of emerging technologies on IPR.

Technology	IPR Challenge	Cybersecurity Risk	Possible Protection
Generative AI			
Artificial Intelligence			
Cloud Computing			
Blockchain			
Social Media			
Digital Platforms			
Software Applications			
Big Data			

Students must complete **at least 5 rows** independently.

Task 10 – Practical IP India Activity

Visit the official IP India website.

Explore:

- Patent search
- Trademark search
- Copyright resources
- Patent e-filing
- Trademark e-filing

IP India provides public-search and e-filing facilities for patents, trademarks, designs and other IPR-related services. ([IP India](https://www.ipindia.gov.in))

Students must record:

- 1. Website visited:**
- 2. Service explored:**
- 3. Purpose of the service:**
- 4. What information can be searched/submitted?**
- 5. How can this service help a technology company?**

Do not submit or file any actual application. This is only an educational exploration activity.

Task 11 – Cyber Incident Response

Assume CyberAI discovers that its source code has been stolen.

Prepare a response using:

Detect

How will the company identify the breach?

Analyse

What logs, accounts, devices and files will be examined?

Contain

How will the company stop further access?

Report

Who should be informed?

Preserve Evidence

What digital evidence should be preserved?

Recover

How will systems and accounts be restored?

Prevent

What controls should be implemented to prevent recurrence?

Task 12 – Final Question

Write **200–250 words** on:

“How can India protect Intellectual Property in an increasingly digital and AI-driven economy through a combination of IPR protection and cybersecurity?”

Your answer must include:

IP India + I4C + Copyright + Patent + Trademark + Trade Secret + Cybersecurity + AI

Final Student Deliverable

Students should prepare the assignment in this order:

- 1. Introduction to IPR**
- 2. Four Major Types of IPR**
- 3. CyberAI Case Study**
- 4. Copyright Analysis**
- 5. Patent Analysis**
- 6. Trademark Analysis**
- 7. Trade Secret & Cybersecurity**
- 8. I4C Analysis**
- 9. IP India vs I4C**
- 10. Emerging Digital Technologies**
- 11. IP India Practical Exploration**
- 12. Cyber Incident Response Plan**
- 13. Conclusion**
- 14. References**

Student Self-Check

- I understand the four major IPR concepts.
- I analysed the CyberAI case.
- I used the official IP India website.
- I explored at least one IP India service.
- I researched I4C.
- I understand the difference between IP protection and cybercrime response.
- I connected cybersecurity with trade-secret protection.
- I analysed emerging digital technologies.
- I prepared an incident-response plan.
- I included official references.

Useful official resources:

[IP India – Patents, Trademarks, Copyright & IPR Services](#)

[I4C – Indian Cyber Crime Coordination Centre](#)